

9. Reflection Questions

Answer these in your lab journal. Be honest and specific.

Part 1: Concepts

1. **Explain the difference between classification, detection, and segmentation in your own words. Give one workforce example of each.**

Classification – assigns a label to an entire image

A quality control system classifying a product as defective or not defective.

Detection – draws a bounding box around objects inside an image

A warehouse camera detects shelving, forklifts and workers to avoid collisions.

Segmentation – only outlines the exact shape of the object at the pixel level inside an image

A medical imaging tool segmenting the outline of a tumor.

2. **What is Non-Maximum Suppression, and why does YOLO use it?**

Non-Maximum Suppression removes duplicate overlapping boxes that all refer to the same object. YOLO predicts lots of boxes but only keeps the highest-confidence box and eliminates the rest.

3. **In your own words, what does IoU measure?**

Intersection of Union means that if there is a perfect overlap, the overlap will be measured at 1 and if there is 0 overlap the overlap will be measured at 0. All numbers between 0 and 1 will be an indication of some level of overlap.

Part 2: Hands-on experience

4. **When you ran YOLO11 on your own image, did it detect everything you expected? Did it detect anything that was wrong? Why do you think it succeeded or failed?**

It detected most of what I expected, but it missed the smaller details like fingers. It succeeded in creating a bounding box that captured the main subject, but, with tiny objects. It struggled with the smaller items possibly because it hadn't been trained with enough similar examples.

5. **When you compared detection and segmentation side by side, name one situation where the box would be enough and one situation where you would really need the mask.**

A box is enough when you are counting people in a room.

A mask is needed to check if a worker is actually wearing safety goggles over their eyes. Boxes only look for an object and the mask can look at the object at the pixel level.

6. SAM 2 does not output category labels. Why is that an important difference from YOLO?

SAM is segmenting shapes and does not care what the shapes are. YOLO handles what the object is and SAM handles exactly where it is. YOLO gets you in the ballpark and SAM gets you to home plate. The models complement each other.

Part 3: Trade-offs

7. Confidence threshold trade-off. Give one application where you would want a low threshold (more sensitive) and one application where you would want a high threshold. Explain why.

Low threshold for more sensitivity is required in medical testing, such as cancer screening – better to over diagnose and retest than to miss detecting cancer.

High threshold for more exact matches would be useful in product recommendations. Recommending a tractor instead of a sports car would not fit most consumer's needs when shopping online for a car. Recommending a similar sports car of a different model or even brand would work, but the tractor would not as it is the wrong product!

8. Precision versus recall. For a workplace safety system that detects unsafe behavior on a factory floor, do you want higher precision or higher recall? What about for a system that tags products in a customer's photo upload?

You want higher recall – missing a dangerous situation is worse than creating a few false alarms.

Higher precision is needed when tagging photos – customers will be annoyed when things are labelled incorrectly. Early computer vision systems struggled with this because the training data wasn't balanced or did not have enough data and sometimes produced highly inaccurate - and at time inappropriate - labels.

9. When would you reach for a specialist model like YOLO11-seg, and when would you reach for a foundation model like SAM 2?

YOLO11-seg is when you want to produce real-time segmentation with known categories. This model is fast and category aware.

SAM 2 is when you need very accurate masks – including objects that the model has never seen. This model is extremely precise and can versatile.

Part 4: Midterm project

10. What project area did you pick for your midterm? In one paragraph, describe: (a) what the application does, (b) where the data will come from, and (c) how you might extend it for the capstone.

I chose to build a PPE Compliance Checker. It is an application that will detect whether workers are wearing their required safety equipment such as glasses, helmets, vests, oxygen tank, etc.

The data will come from publicly available PPE datasets on Kaggle, Roboflow, and any images that I can capture of people either wearing or not wearing the PPE.

For capstone, I would like to extend the study into a real time video pipeline with worker tracker, SAM 2 segmentation refinement, persistent compliance monitoring, and an alerting dashboard that mirrors real industrial safety systems.